Convergence Simulations Model B

Vera Schroeder

October 2026

1 Model Structure

1.1 Data and indexing

The data are N binary trials. Trial i has

- a response $y_i \in \{0, 1\}$, the monkey's categorization of the stimulus as "low" (0) or "high" (1);
- a standardized stimulus amplitude x_i. The raw amplitudes are centered and scaled by the mean and standard deviation of the distinct amplitude levels. The covariate row is $(1, x_i)$;
- a group $g[i] \in \{1, \ldots, 4\}$: the combination of hand and distractor condition,

  $$g \in \{\text{left_uni},\ \text{left_bi},\ \text{right_uni},\ \text{right_bi}\},$$

  where "uni" is unimanual (no distractor) and "bi" is bimanual (distractor present);
- a session $s[i] \in \{1, \ldots, S\}$; the real data have $S = 42$.

1.2 Likelihood

Each response is Bernoulli,

$$y_i \sim \text{Bernoulli}(\psi_i), \qquad \psi_i = \psi\big(x_i;\ \theta_{g[i],s[i]}\big),$$

with the psychometric function

$$\psi(x; \theta) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x), \qquad \theta = (\beta_0, \beta_1, \gamma_h, \gamma_l).$$

Here β_0 and β_1 are parameters of the sigmoid, and γ_h, γ_l set the asymptotes:

$$\lim_{x \to -\infty} \psi = \gamma_h, \qquad \lim_{x \to +\infty} \psi = 1 - \gamma_l.$$

So γ_h is the probability of responding 1 ("high") at very low amplitudes, and γ_l the probability of responding 0 ("low") at very high amplitudes. In the code ψ_i is clipped to $[10^{-8}, 1 - 10^{-8}]$ before entering the Bernoulli likelihood, for numerical stability.

1.3 Session-level

For each group g and session s, the four parameters are drawn independently from group-specific distributions:

$$\begin{aligned}
\beta_{0,gs} &\sim \mathcal{N}\big(\mu_{\beta_0,g},\ \sigma_{\beta_0,g}\big),\\
\beta_{1,gs} &\sim \mathcal{N}\big(\mu_{\beta_1,g},\ \sigma_{\beta_1,g}\big),\\
\gamma_{h,gs} &= 0.25\, w_{h,gs}, \qquad w_{h,gs} \sim \text{LogitNormal}\big(\mu_{\gamma_h,g},\ \sigma_{\gamma_h,g}\big),\\
\gamma_{l,gs} &= 0.25\, w_{l,gs}, \qquad w_{l,gs} \sim \text{LogitNormal}\big(\mu_{\gamma_l,g},\ \sigma_{\gamma_l,g}\big).
\end{aligned}$$

The LogitNormal keeps $w \in (0, 1)$, so each lapse parameter γ lies in $(0, 0.25)$. That range keeps the curve increasing and makes the PSE and JND below well defined.

Groups are independent of each other (no pooling across groups).

1.4 Group-level Hyperpriors

Each of the four groups has its own 8 hyperparameters, with the same priors in every group:

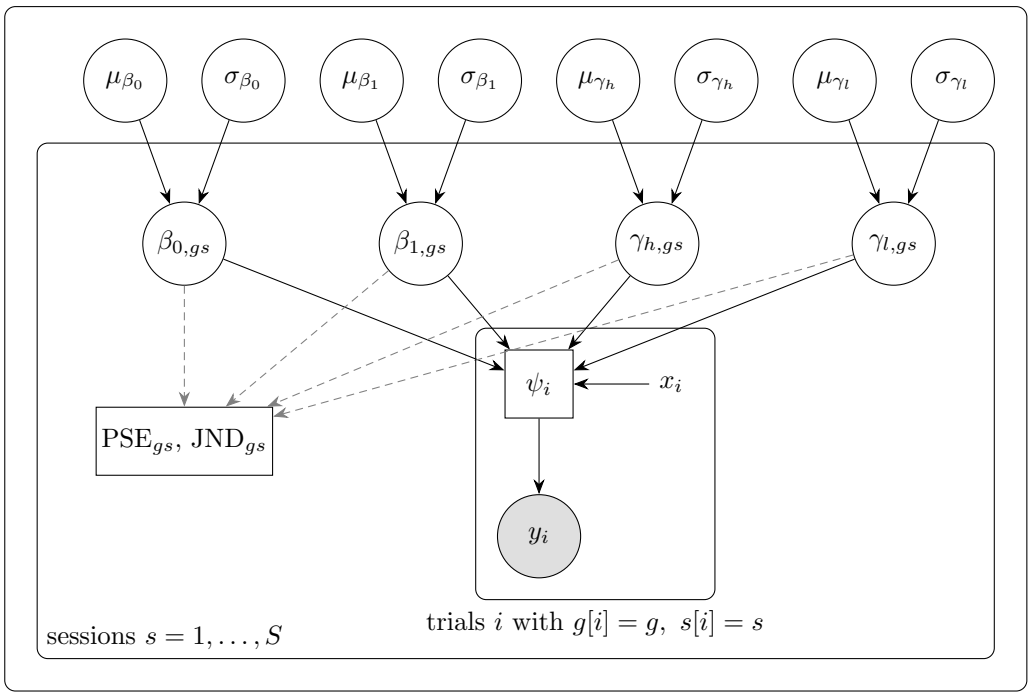
$$\begin{aligned}
\mu_{\beta_0,g} &\sim \mathcal{N}(0,\ 2), & \sigma_{\beta_0,g} &\sim \text{HalfNormal}(2),\\
\mu_{\beta_1,g} &\sim \text{LogNormal}(\log 4,\ 0.5), & \sigma_{\beta_1,g} &\sim \text{Exponential}(\lambda = 1) \quad \text{(mean 1)},\\
\mu_{\gamma_h,g},\ \mu_{\gamma_l,g} &\sim \mathcal{N}(-3,\ 1.5), & \sigma_{\gamma_h,g},\ \sigma_{\gamma_l,g} &\sim \text{HalfNormal}(0.5).
\end{aligned}$$

All hyperparameters are independent. The priors were chosen from the stimulus design and general knowledge of the task, and checked by prior predictive simulation (`Check_Prior_B.py`), not from fits to the real data:

- μ_{β_0}: the stimulus amplitudes (6 to 50) are symmetric around 28, the category boundary, which standardizes to $x = 0$. This prior puts a group's typical PSE between amplitudes 13 and 43 with 90% probability, and outside the stimulus range only 3% of the time.
- σ_{β_0}: weakly informative, bounded by the design. The implied day-to-day sd of the PSE has median about 5 amplitude units and is below about 20 with 95% probability, small next to the stimulus range.
- μ_{β_1}: positive with no upper bound; median 4, 95% in [1.5, 10.6].
- σ_{β_1}: mean 1, so that sessions vary around the group slope without the session slopes turning negative (2% prior probability).
- μ_γ (logit scale): the typical lapse rate $0.25\ \text{logit}^{-1}(\mu_\gamma)$ has median 0.012 and 95% range [0.0007, 0.12]; trained animals lapse on a few percent of trials and rarely more than about 10%.
- σ_γ (logit scale): median 0.34. For a group whose typical lapse rate is 0.05, this puts about 80% of sessions between 0.03 and 0.07.

An earlier version of the model used $\mu_{\beta_0} \sim \mathcal{U}[-12,\ 12]$, $\sigma_{\beta_0} \sim$ Exponential(mean 12), $\mu_{\beta_1} \sim \mathcal{U}[0,\ 8]$, $\sigma_{\beta_1} \sim$ Exponential(mean 4), $\mu_\gamma \sim \mathcal{N}(-2.5,\ 1)$ and $\sigma_\gamma \sim$ HalfNormal(1.25). Its convergence study motivated the changes; Section 3.6 compares the two versions.

1.5 Graphical Model



Circles are random variables (shaded: observed), rectangles are deterministic functions, and x_i is a fixed covariate. Dashed arrows lead to derived quantities that do not feed back into the likelihood.

1.6 Derived Quantities: PSE and JND

For each group and session the model records two summaries of the psychometric curve.

Point of subjective equality (PSE): the stimulus at which a response of "high" and "low" are equally likely ($\psi = 1/2$). Solving $\psi(x;\theta) = \frac{1}{2}$ for x gives

$$\text{PSE} = \frac{1}{\beta_1}\left[-\beta_0 + \log\frac{1-2\gamma_h}{1-2\gamma_l}\right].$$

Just noticeable difference (JND): the amount the stimulus must be changed for the difference to be noticeable half of the time, a measure of sensitivity. This is computed as half the stimulus distance between the points where $\psi = 0.25$ and $\psi = 0.75$,

$$\text{JND} = \frac{x_{.75} - x_{.25}}{2} = \frac{1}{2\beta_1}\,\log\frac{(3-4\gamma_h)(3-4\gamma_l)}{(1-4\gamma_h)(1-4\gamma_l)}.$$

Both are in units of the standardized stimulus. When $\gamma_h = \gamma_l = 0$, they reduce to $\text{PSE} = -\frac{\beta_0}{\beta_1}$ and $\text{JND} = \frac{\log 3}{\beta_1}$.

1.7 Implementation

To aid fit, as in the convention of typical logistic regression, the raw amplitudes x_i are centered and scaled by the mean and standard deviation of the distinct amplitude levels. To aid convergence of HMC NUTS, all distributions are written as deterministic transforms of a standard normal or uniform random variable. These non-centered parameterizations mean the sampler only works with uniform or standard normal variables.

2 Synthetic Data Generation

I fixed "hyperparameters" and simulated 10 total dataset conditions in two sweeps, and fit the model to each dataset. Each sweep was performed twice with new data. The true hyperparameters are the dashed lines in the recovery plots below; in particular, the true group slopes are $\mu_{\beta_1} = 3$ for the left hand and 6 for the right hand.

The first sweep held trials-per-session constant and varied number of sessions (5, 10, 20, 40 and 80 sessions, each with 308 trials). This provides info on convergence of global hyperparameters.

The second held session count constant, varying trials-per-session (42 sessions, each with 20, 40, 80, 160 or 320 trials). This provides info on convergence of session-level parameters.

The real data has 42 sessions with about 308 trials per session, so the 40-session and 320-trial fits are the most representative.

Both versions of the model (current and earlier priors) were fit to exactly the same synthetic datasets, so any difference between their results comes from the priors alone. The results below are for the current priors; Section 3.6 compares the two.

3 Model Fit Results

Every figure and number in this report is produced by `Convergence_B.py`; the cell "Numbers for Convergence_B_Report.tex" prints the numbers that the earlier cells do not, and also runs on the saved results without refitting. The statements about what the priors imply come from `Check_Prior_B.py`. Session-level RMSEs are pooled over both replicates (84 sessions per group in the trials sweep).

3.1 Sampling Diagnostics

All 20 fits sampled without divergences, with $\hat{R} \leq 1.010$ and a bulk ESS of at least 579 for every hyperparameter. This includes the 5-session fits, which had 94 divergences under the earlier priors (93 of them in one replicate) and had to be excluded.

Table 1: Sampler diagnostics over both replicates. $\hat{R}$ and bulk ESS are taken over the hyperparameters.

Sweep	Size	max $\hat{R}$	min ESS	divergences
sessions	5	1.005	1249	0
sessions	10	1.005	1253	0
sessions	20	1.004	1202	0
sessions	40	1.006	1018	0
sessions	80	1.005	928	0
trials	20	1.008	1018	0
trials	40	1.010	579	0
trials	80	1.004	675	0
trials	160	1.005	690	0
trials	320	1.004	1033	0

3.2 Hyperparameter means

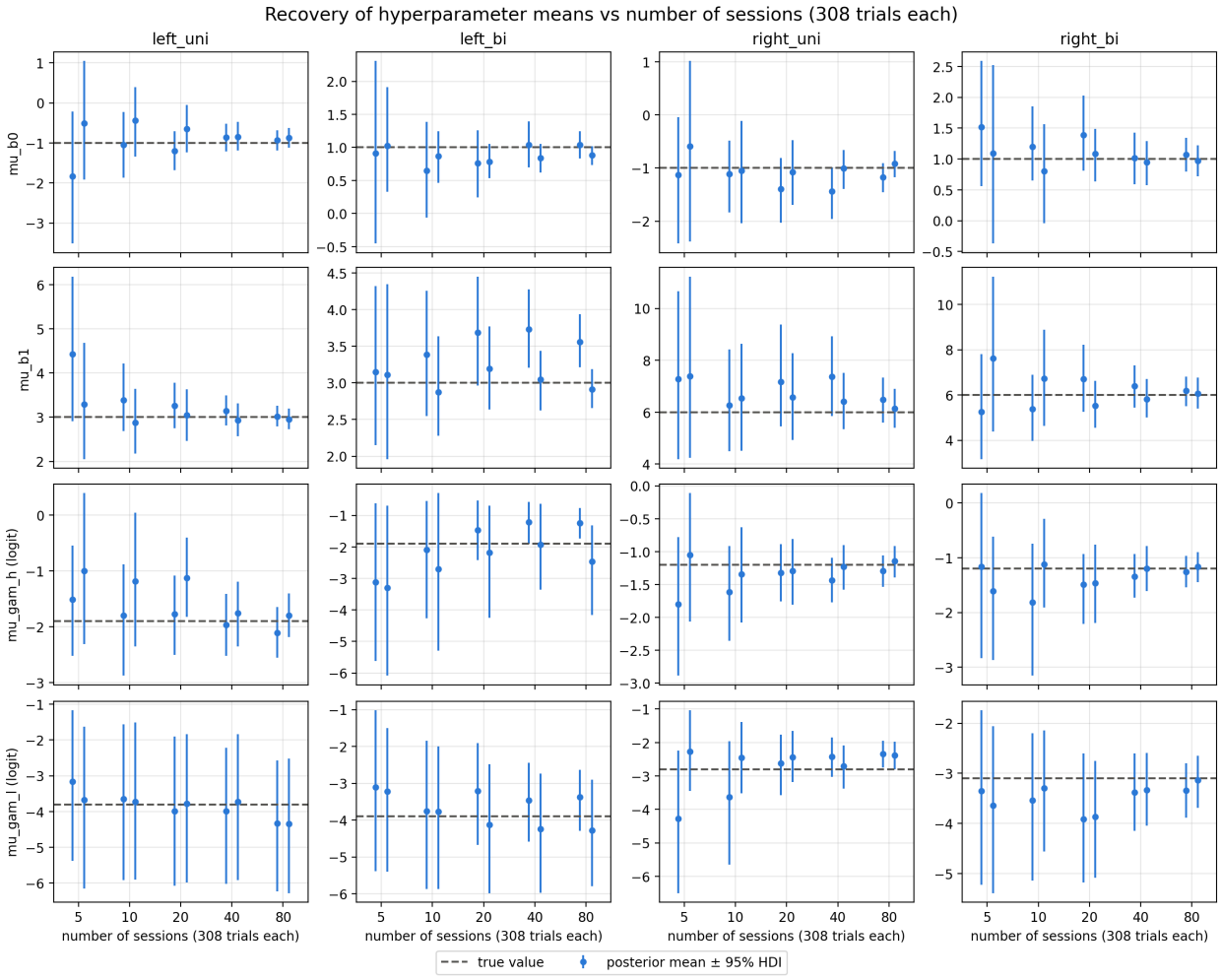


Figure 1: Posterior mean and 95% HDI of the group means against the number of sessions (308 trials each). The two replicates are drawn side by side; the dashed line is the true value.

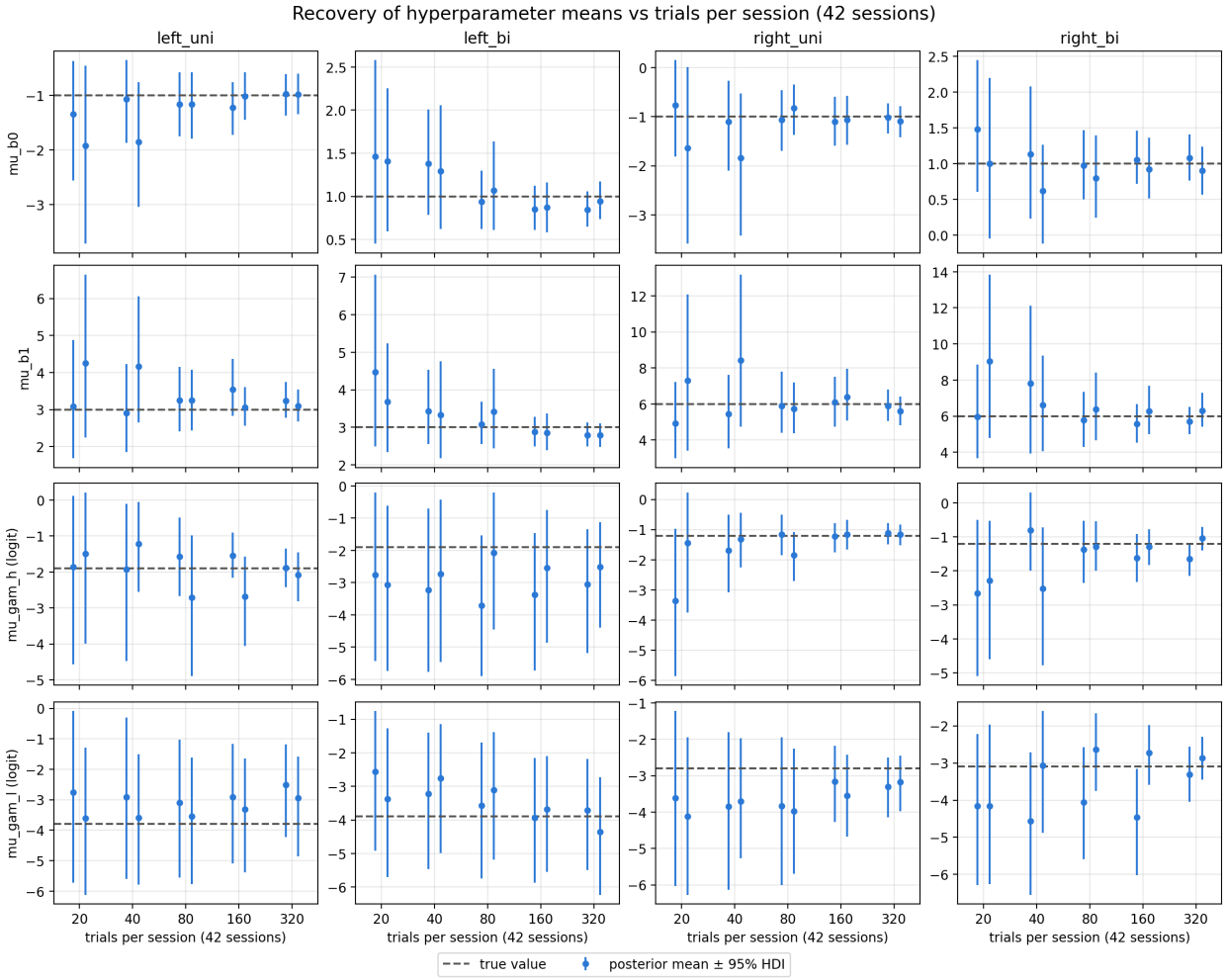


Figure 2: As Figure 1, against trials per session (42 sessions).

Bias μ_{β_0}: Recovered well in all groups. At 40 sessions the posterior means (averaged over replicates) are within 0.22 of the truth, with HDI widths of 0.56–0.85.

Slope μ_{β_1}: For the left-hand groups (true value $\mu_{\beta_1} = 3$), the posterior converges to the truth (at 80 sessions: 2.98 [2.76, 3.23] and 3.24 [2.93, 3.56]). For the right-hand groups (true value 6) the posterior is much less precise: at 40 sessions right_uni is 6.89 [5.60, 8.22] and right_bi 6.11 [5.23, 7.01], and at 80 sessions 6.32 [5.50, 7.11] and 6.13 [5.45, 6.79]. The posteriors are no longer truncated at 8, as they were under the earlier bounded prior; the HDI of right_uni now reaches up to about 8.9 at 40 sessions in one replicate.

The main cause is the stimulus design rather than the prior. At a slope of 6 the rise from $\psi = 0.25$ to $\psi = 0.75$ spans $2 \log 3/6 \approx 0.37$ standardized units, less than the spacing between stimulus levels (about 0.41), so only the two levels nearest the PSE carry information about the slope; at the other levels the curve is already at its floor or ceiling. The data can therefore rule out shallower slopes but not steeper ones, so the right-hand posteriors are wide and, in the sessions sweep, lean slightly above the truth. Only stimulus levels spaced more closely around the PSE would make steep slopes better identified.

Lapse-rate means μ_{γ_h} and μ_{γ_l}: The lapse rates are only identifiable at stimulus levels where the sigmoid has reached its floor or ceiling, and lapses are rare there (about 2 expected lapses of γ_l per stimulus level for the left-hand groups at the real data size).

μ_{γ_h} is recovered for left_uni and both right-hand groups (HDI widths 0.7–1.1 at 40 sessions). For left_bi, whose sigmoid never reaches its floor within the stimulus range, the posterior is wide and in the trials sweep sits below the truth: -2.79 [-4.79, -1.23] at 320 trials (true -1.9).

μ_{γ_l} for the left-hand groups is now recovered without bias in the sessions sweep: at 40 sessions -3.86 for both (true -3.8 and -3.9), although the HDIs are wide (2.7–3.9). In the trials sweep left_uni is still too high, -2.73 [-4.55, -1.38] at 320 trials. The left_uni sigmoid only reaches 0.97 at the highest stimulus, so a shortfall at the top can be explained by γ_l or by the slope and bias, and with so few lapses the estimate depends strongly on the particular dataset. For right_uni the HDI at 80 sessions excludes the true value in both replicates (-2.36 [-2.77, -1.96], true -2.8). The estimate was just as high under the earlier priors (-2.37, with the HDI excluding the truth in one of the two replicates), so it is not caused by the prior, and its cause has not been identified.

3.3 Hyperparameter standard deviations

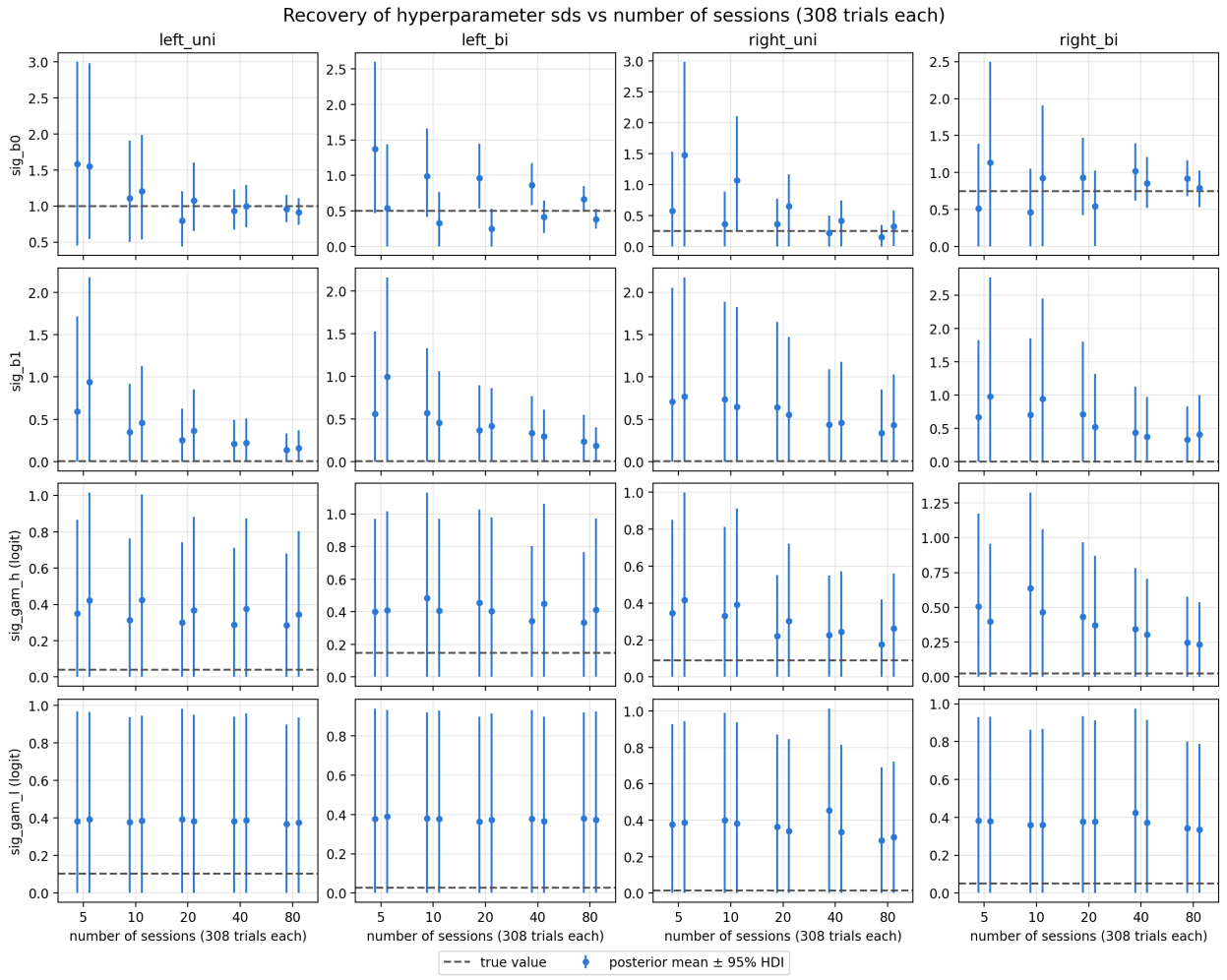


Figure 3: Posterior mean and 95% HDI of the between-session standard deviations against the number of sessions.

Table 2: Between-session sds at 80 sessions: true value and posterior mean [95% HDI], averaged over replicates.

	left_uni	left_bi	right_uni	right_bi
σ_{β_0} true	1.00	0.50	0.25	0.75
posterior	0.94 [0.76, 1.13]	0.52 [0.37, 0.69]	0.24 [0.00, 0.46]	0.85 [0.60, 1.10]
σ_{β_1} true	0.006	0.004	0.008	0.002
posterior	0.15 [0.00, 0.35]	0.21 [0.00, 0.48]	0.39 [0.00, 0.94]	0.37 [0.00, 0.92]
σ_{γ_h} true	0.040	0.147	0.090	0.024
posterior	0.31 [0.00, 0.74]	0.37 [0.00, 0.87]	0.22 [0.00, 0.49]	0.24 [0.00, 0.56]
σ_{γ_l} true	0.103	0.027	0.014	0.050
posterior	0.37 [0.00, 0.92]	0.38 [0.00, 0.92]	0.30 [0.00, 0.71]	0.34 [0.00, 0.79]

σ_{β_0} is recovered at 80 sessions (Table 2), and its HDIs contain the true value in 88–100% of fits at every size. With fewer sessions the estimates can still be well off: at 20 sessions right_uni is 0.51 against a true 0.25. The other three sds, σ_{β_1}, σ_{γ_h} and σ_{γ_l}, have true values near zero, meaning the simulated sessions barely differ in slope or lapse rate. Their HDIs contain the true value but stay wide, so these sds cannot be recovered at these data sizes. For σ_{γ_l} the posterior is essentially the prior at every size (posterior means about 0.3–0.4 and HDIs up to about 0.7–1.0, against a prior mean of 0.40 and 95% point of 0.98): the data carry almost no information about how much γ_l varies between sessions.

3.4 Session-level parameters

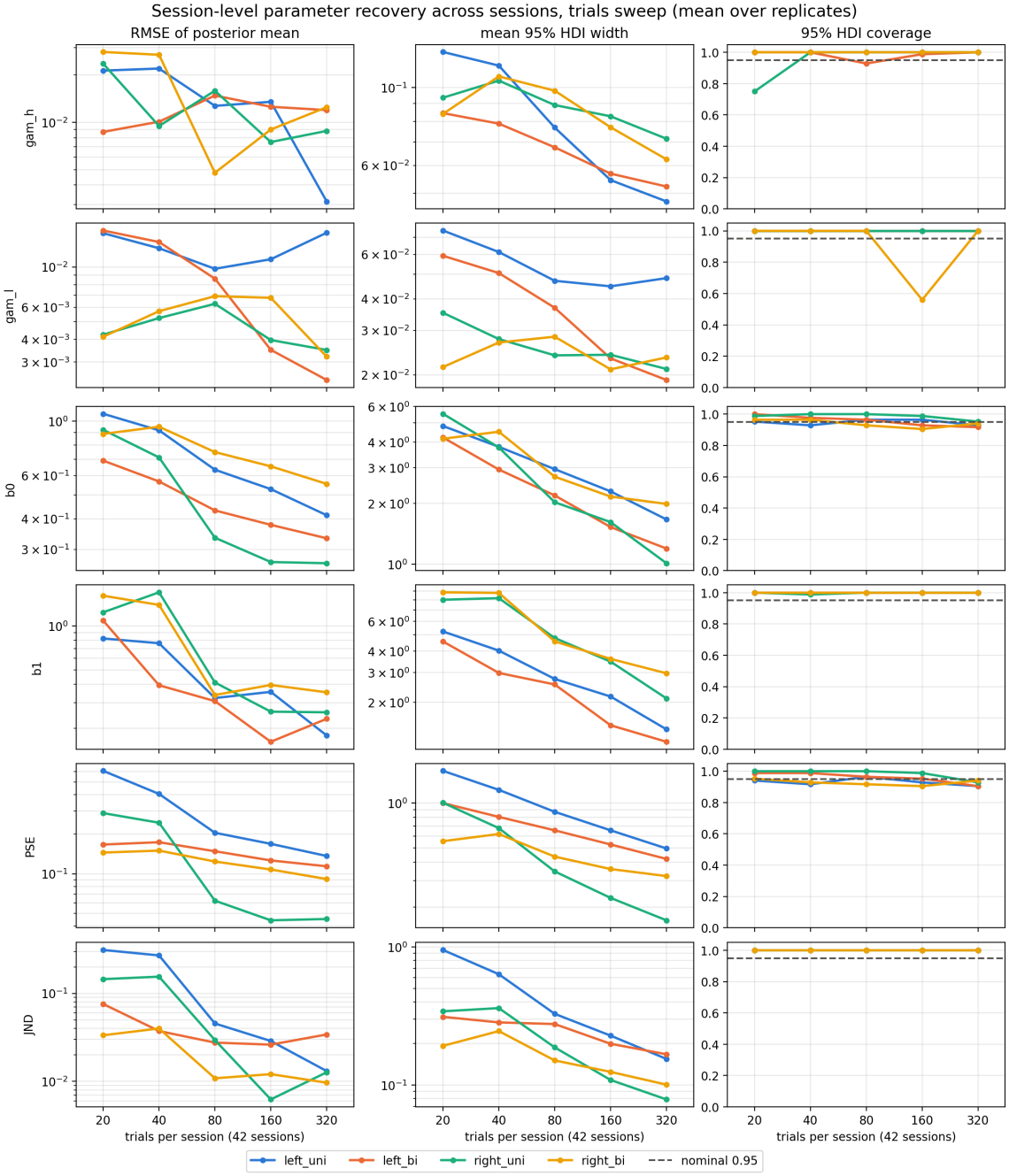


Figure 4: Session-level recovery against trials per session (42 sessions), averaged over replicates. Left: RMSE of the posterior mean across sessions. Middle: mean 95% HDI width. Right: fraction of sessions whose HDI contains the true value.

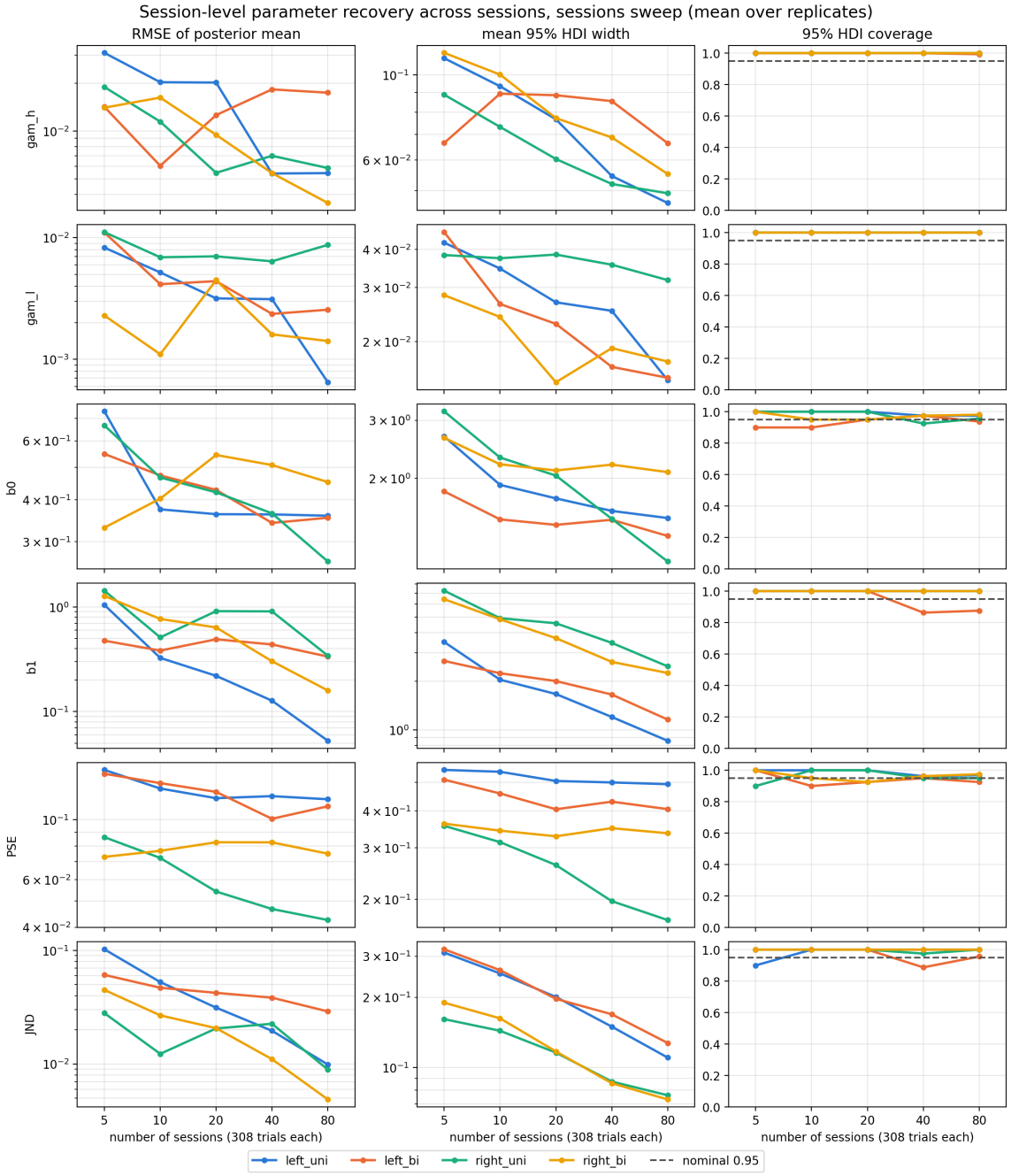


Figure 5: As Figure 4, against the number of sessions (308 trials each).

Table 3: Session-level recovery at 320 trials per session (42 sessions, both replicates): RMSE of the posterior mean / mean HDI width / coverage.

	left_uni	left_bi	right_uni	right_bi
PSE	0.14 / 0.49 / 0.90	0.12 / 0.42 / 0.90	0.05 / 0.16 / 0.93	0.09 / 0.32 / 0.94
JND	0.014 / 0.15 / 1.00	0.034 / 0.17 / 1.00	0.013 / 0.08 / 1.00	0.010 / 0.10 / 1.00
β_0	0.41 / 1.67 / 0.93	0.34 / 1.20 / 0.92	0.26 / 1.01 / 0.95	0.55 / 1.99 / 0.94
β_1	0.19 / 1.39 / 1.00	0.23 / 1.18 / 1.00	0.29 / 2.11 / 1.00	0.35 / 2.97 / 1.00
γ_h	0.003 / 0.047 / 1.00	0.012 / 0.052 / 1.00	0.009 / 0.071 / 1.00	0.013 / 0.062 / 1.00
γ_l	0.016 / 0.048 / 1.00	0.003 / 0.019 / 1.00	0.004 / 0.021 / 1.00	0.003 / 0.023 / 1.00

PSE, JND, β_0, β_1. Error and HDI width fall steadily as trials per session grow (Figure 4); at 320 trials the RMSE of the PSE is 0.05–0.14 and of the JND 0.01–0.03 (Table 3). The largest drop is between 40 and 80 trials; below 80 trials the JND in particular is poorly determined for some groups (RMSE 0.40 and 0.34 for left_uni at 20 and 40 trials, 0.19 for right_uni), falling to 0.05 or less at 80 trials. Adding sessions at a fixed 308 trials (Figure 5) also improves the session estimates, because better-estimated hyperparameters give better shrinkage.

Coverage. The coverage of β_1 and JND is essentially 1.0: the HDIs are wider than needed, because the between-session sd of the slope is overestimated. PSE coverage at 320 trials is slightly below nominal (0.90–0.94). With 84 sessions per group, the standard error of a coverage estimate near 0.95 is about 0.024, so 0.90 is about 2 standard errors low: a mild undercoverage worth keeping in mind. Errors at the population level carry through to every session of a group: at 40 sessions, one replicate's upward error in μ_{β_1} for left_bi (3.73 against a true 3) lowers the coverage of β_1 and JND in that group to 0.72 and 0.78 in that replicate (0.86 and 0.89 over both replicates).

Lapse rates γ_h and γ_l. The tighter prior on σ_γ shrinks the session lapse rates more strongly toward their group's value, so their HDIs are narrower than under the earlier priors (mean width at 320 trials 0.058 for γ_h and 0.028 for γ_l). Coverage is mostly 1.0, but a miss at the group level now shows up in every session of the group: at 160 trials, one replicate's right_bi group γ_l was estimated at 0.004 against a true 0.011, which lowers the overall γ_l coverage at that size to 0.89.

For left_uni, γ_l remains too high in the trials sweep: at 320 trials the posterior means average 0.021 against a true 0.0056 (RMSE 0.016). This is the μ_{γ_l} error described above passed down to every session. In the sessions sweep the same quantity is recovered well (RMSE 0.003 at 40 sessions). For left_bi, γ_h is too low in the trials sweep (0.021 against 0.032 at 320 trials), the session-level counterpart of the μ_{γ_h} error.

3.5 Posterior densities

The following plots show the same fits as full posterior densities, darker for larger datasets. They confirm the points above: the μ_{β_1} posteriors of the right-hand groups are wide but no longer cut off at 8, and the posteriors of σ_{β_1}, σ_{γ_h} and especially σ_{γ_l} are broad and change little with the amount of data.

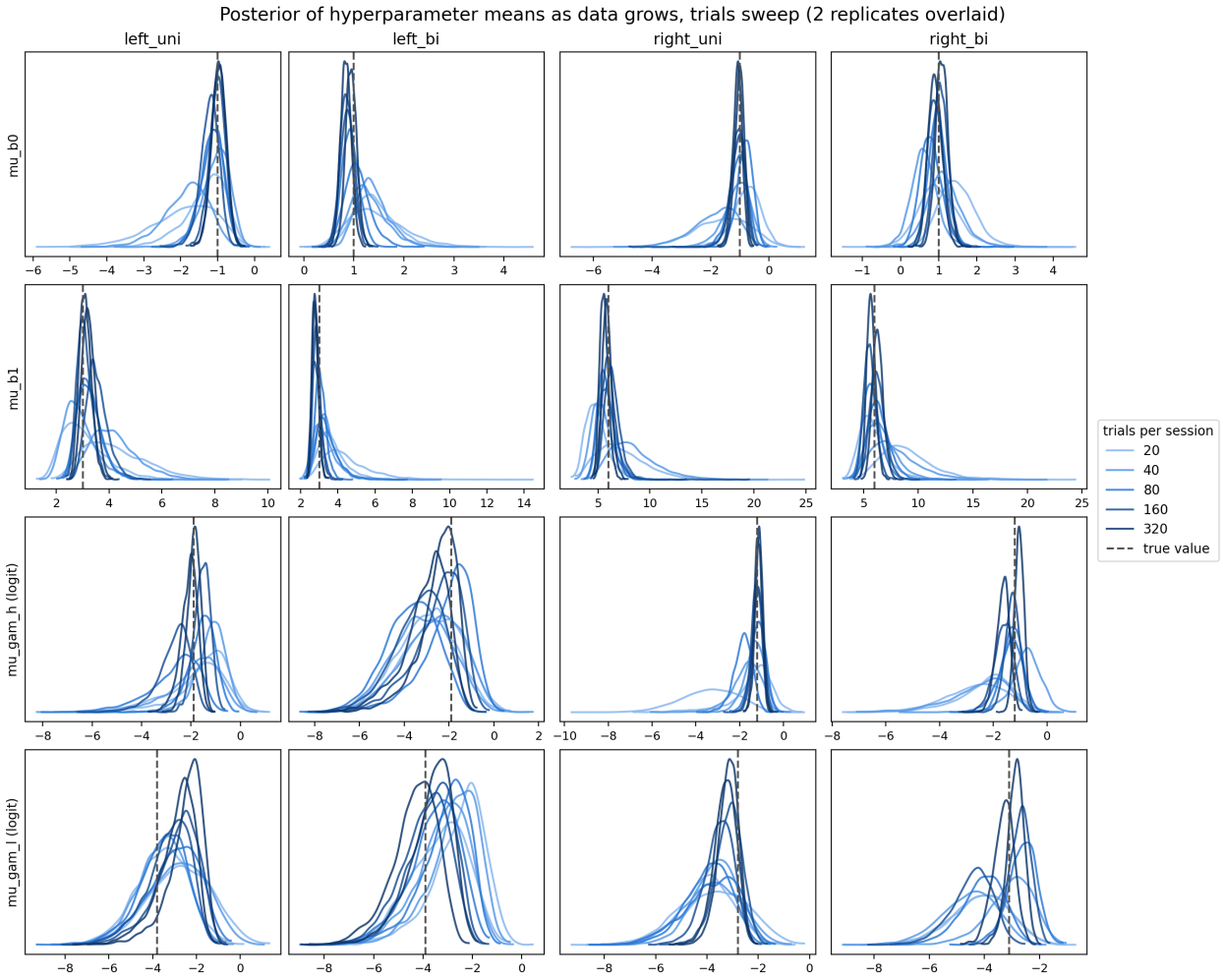


Figure 6: Posteriors of mean hyperparameters as trials vary

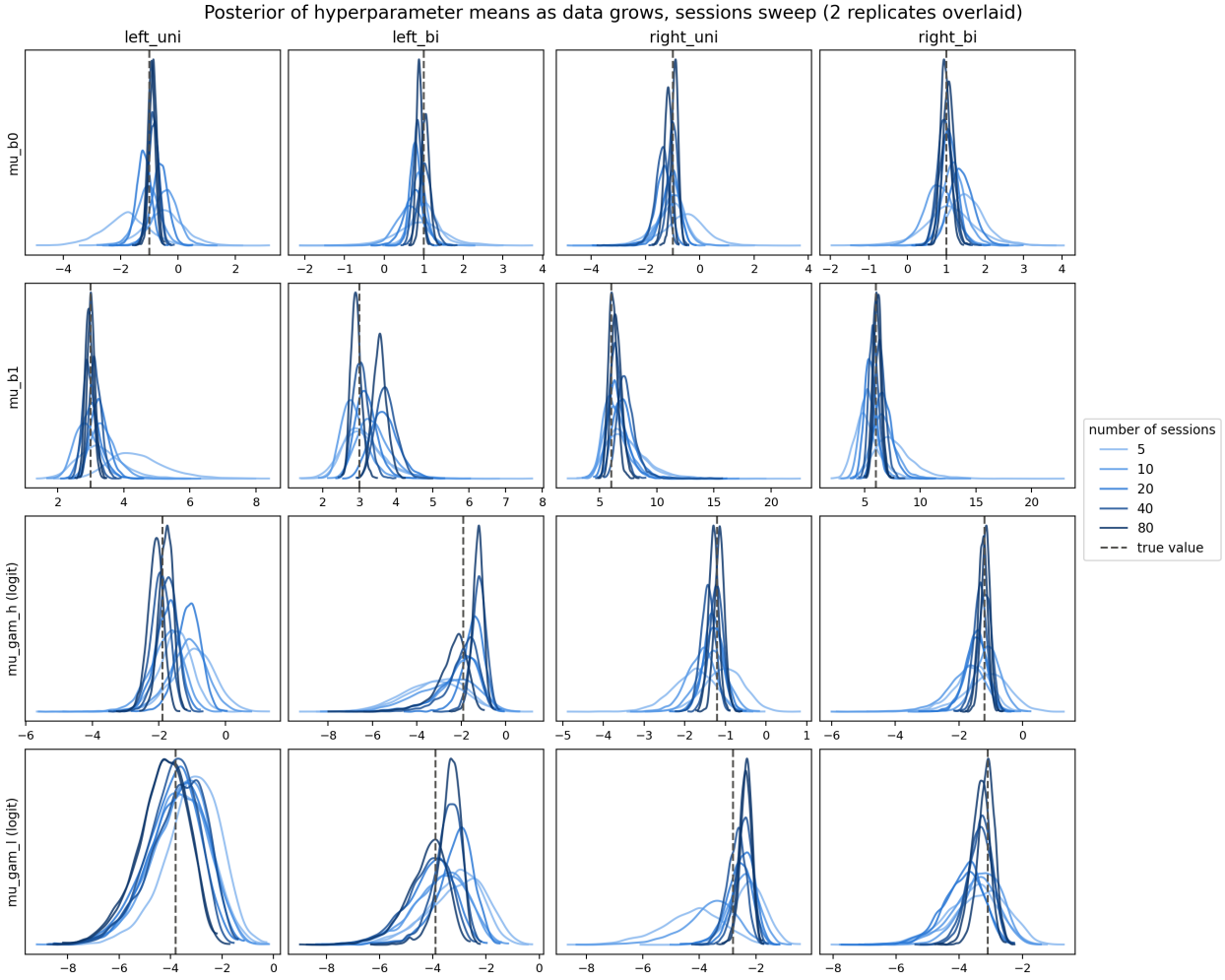


Figure 7: Posteriors of mean hyperparameters as sessions vary

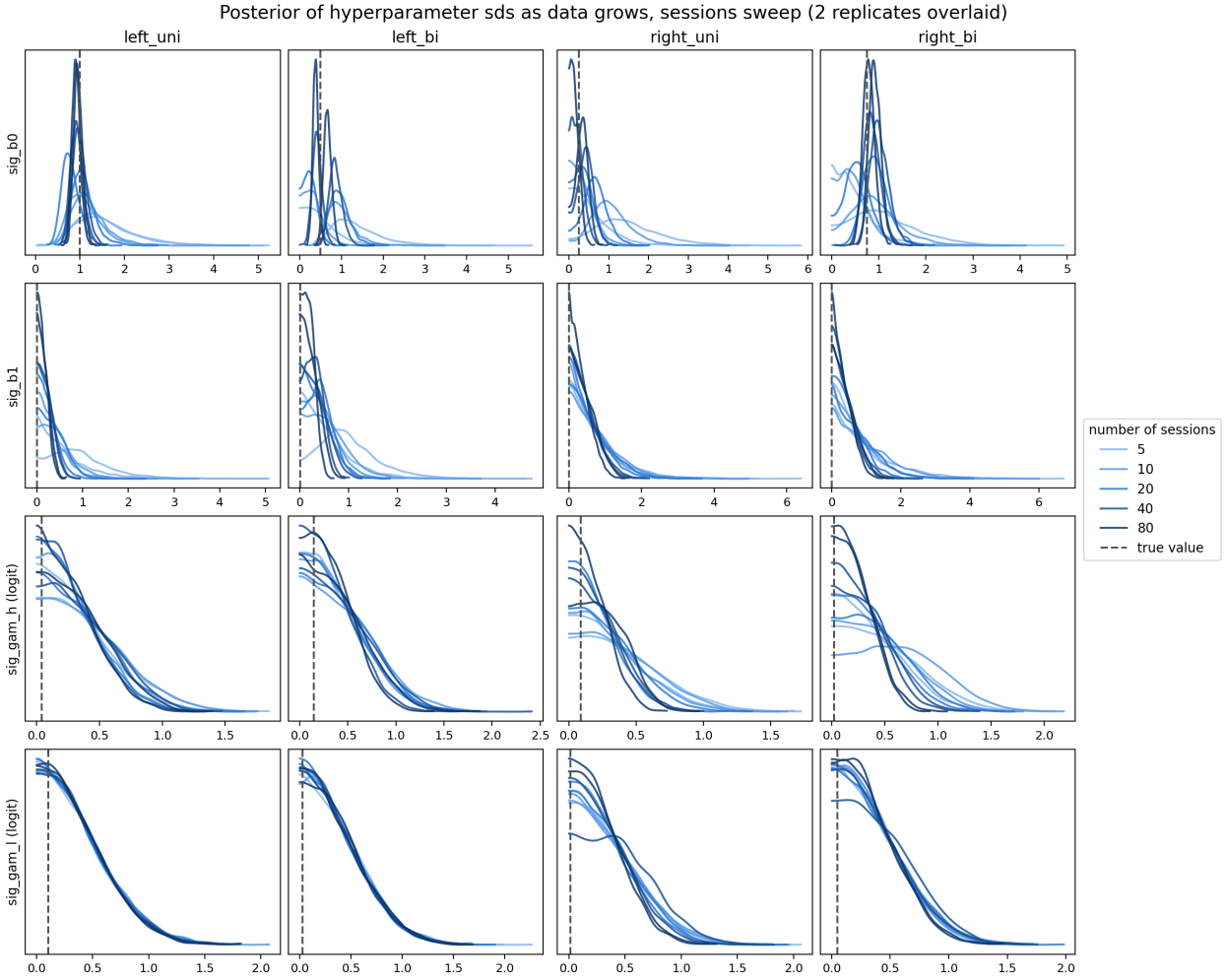


Figure 8: Posteriors of sd hyperparameters as sessions vary

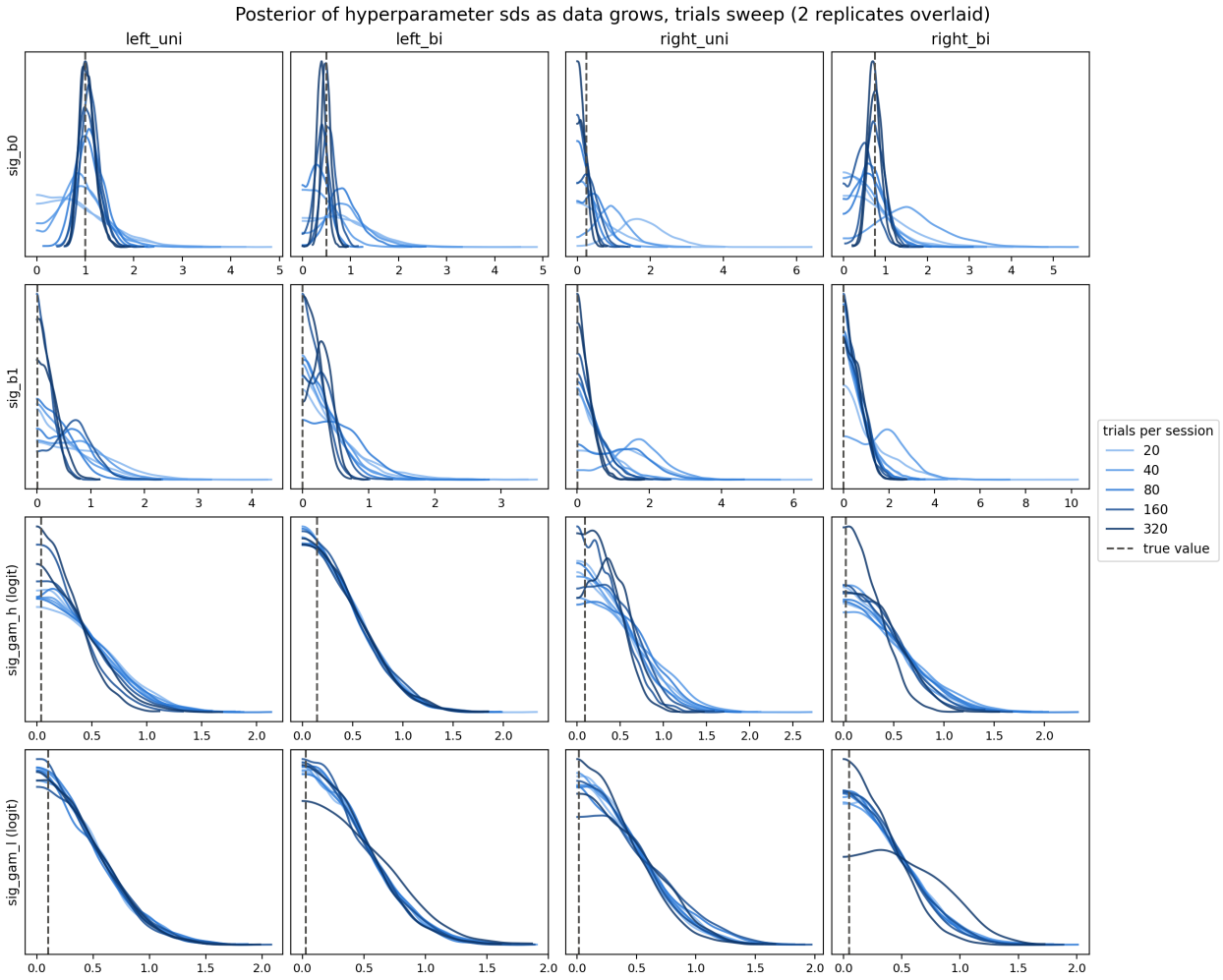


Figure 9: Posteriors of sd hyperparameters as trials vary

3.6 Comparison with the earlier priors

Both versions were fit to the same synthetic datasets, so Table 4 isolates the effect of the prior changes.

Table 4: Earlier vs current priors on the same synthetic datasets. Hyperparameter entries are posterior mean [95% HDI], averaged over replicates.

	true	earlier priors	current priors
5 sessions: div. / max $\hat{R}$ / min ESS		94 / 1.061 / 77	0 / 1.005 / 1249
other fits: min ESS		431–867	579–1253
μ_{β_1} right_uni, 40 sessions	6	6.88 [5.93, 7.85]	6.89 [5.60, 8.22]
μ_{β_1} right_bi, 80 sessions	6	6.21 [5.53, 6.92]	6.13 [5.45, 6.79]
μ_{γ_l} left_uni, 40 sessions	−3.8	−3.30 [−4.70, −2.02]	−3.86 [−5.97, −2.03]
μ_{γ_l} left_bi, 40 sessions	−3.9	−3.57 [−4.67, −2.57]	−3.86 [−5.28, −2.60]
μ_{γ_l} left_uni, 320 trials	−3.8	−2.61 [−3.92, −1.46]	−2.73 [−4.55, −1.38]
γ_l left_uni, session RMSE, 40 sessions		0.0087	0.0034
σ_{γ_l} posterior means, 80 sessions	0.01–0.10	0.37–0.65	0.30–0.38
mean HDI width of γ_h / γ_l, 320 trials		0.070 / 0.037	0.058 / 0.028

In summary, the current priors

- remove the sampling problems of the small (5-session) datasets;
- remove the truncation of the right-hand slopes at 8, so their HDIs now honestly reflect how little the design says about steep slopes;
- remove the upward bias of μ_{γ_l} for the left-hand groups in the sessions sweep; the bias remains for left_uni in the trials sweep, where it comes from the curve not reaching its ceiling rather than from the prior;
- narrow the session-level lapse-rate HDIs, at the cost of occasional whole-group misses (Section 3.4).

3.7 Conclusions for the real data

At the size of the real dataset (42 sessions of about 308 trials):

- **Reliable:** per-session PSE and JND; group means μ_{β_0}; μ_{β_1} for the left-hand groups; the bias spread σ_{β_0}; μ_{γ_h} except for left_bi.
- **Weakly constrained:** μ_{β_1} for the right-hand groups, whose HDIs are about 1.8–2.6 units wide at 40 sessions.
- **Biased in some datasets:** lapse rates whose side of the curve does not reach its asymptote within the stimulus range: γ_l for left_uni (estimated too high) and γ_h for left_bi (too low).
- **Not identified:** σ_{β_1}, σ_{γ_h} and σ_{γ_l} when their true values are near zero. Their posteriors largely reflect the priors and should not be interpreted.
- Session-level HDIs are mostly conservative (coverage near or above 95%), with a mild under-coverage for the PSE (0.90–0.94 at 320 trials).

Possible follow-ups.

1. *Done (October 2026):* the real data were refit with the current priors; see the separate report "Model B Fit to the Real Data" (`Fit_B_Report.tex`).

2. Find out why μ_{γ_l} for right_uni is overestimated at 80 sessions in both replicates and under both sets of priors.

3. Consider removing the session-level variation of the lapse rates (one γ_h and γ_l per group), since σ_{γ_l} and σ_{γ_h} are not identified; this would also simplify the model.

4. For future experiments: stimulus levels spaced more closely around the PSE would identify steep slopes, and levels further out would let the curves reach their asymptotes and identify the lapse rates.